GENERAL RISK ASSESSMENT FORM

Ref:

Describe the activity being assessed:	**Assessed by:**	**Endorsed by:**
Remote controlled operation and set-up of the Lynx motion Robotic Arm for test tube pick-n-place movement with the University of West of England, UFMFX8-30-3 class, in Z Block.	Logan Smith - 22020998	Dr. Carlos Insurable (Module Leader)
Who might be harmed: Group I5 Students, Lecturers, UWE SENG Technicians **How many exposed to risk:** 5-10	**Date of Assessment:** 09/10/2025	**Review date:** 09/10/2026

Hazards Identified *(state the potential harm)*	**Existing Control Measures**	**S**	**L**	**Risk Level**	**Additional Control Measures**	**S**	**L**	**Risk Level**	**By whom and by when**	**Date completed**
Exposed Wiring/Circuitry Hazard: Electrical Shock, Fire & Potential Carcinogenic Fume Inhalation	• Most if not all, wiring is heat shrinked • Microcontroller is low voltage (5V) • Supervised during operation • Supervisors have high level of electrical competence • Group I5 has above average level of electrical competence • Arm is controlled from a distance with little need for interaction within range	2	2	4	• New users are given briefing & demonstration before operation • Device is inspected before every operation • Emergency stop switch is accessible during testing • Replace any damaged wiring or connectors when found immediately	2	1	2	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing
Rapid Robotic Arm Movement: Physical Impact, Pinch Points	• Supervised by staff during operation • Supervisors have high level of electro-mechanical competence • Group I5 has above average level of electro-mechanical competence • Arm is controlled from a distance with little need for interaction within range	2	3	6	• Robotic arm is powered off during close interactions • Safe operating zones are indicated before testing (maximum robotic arm reach) • Power plug is accessible • Set slow movement speeds during testing	2	1	2	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing

Power Supply (Battery) Failure: Chemical Burns, Carcinogenic Inhalation	• Arm is controlled from a distance with little need for interaction within range • Battery is enclosed within a housing • Only approved battery types are used	3	2	6	• Battery is inspected before each use to ensure it is not only charged but safe to use • Store and charge batteries in a safe location • Never charge unattended or near flammable materials • Keep fire extinguisher nearby (z-block)	3	1	3	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing
Power Supply (Outlet) Miss Use: Mains Voltage Electrocution, Fire & Potential Carcinogenic Fume Inhalation	• Sockets are fused and meet UK regulations for sockets • Medical Trained Staff in Z-Block • Staff and students trained in electrical safety	4	2	8	• Only use inspected/tested equipment and extension leads • Do not daisy chain extension cords • Disconnect power when not in use	4	1	4	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing
Broken Glass (Test-Tube Movement Failure): Potential Cut Risk, Amplified by Rapid Robotic Arm Movement	• Arm is controlled from a distance with little need for interaction within range • Robotic arm is programmed for this task with smooth motions (for finished product) • Robotic arm is tested with dummy test-tube to meet bare minimum strength requirements for finished product	2	3	6	• Keep area clear of bystanders during any operation • Place protective shield or cover between user and robotic arm when stress testing • Recommended PPE for stress testing: safety glasses and gloves	2	1	2	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing
Trip Hazards (Cables): Slips, Trips, Falls	• Workspace is maintained to a professional standard • Cables are routed safely	2	2	4	• Tape down any loose cables • Do a check of the workspace before and after operation • Keep personal belongings off the operational space	2	1	2	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing
Overheating Components: Burns or fire from overheating parts	• Low voltage circuits generate minimal heat • Components visually inspected before operation	2	2	4	• Allow components to cool between extended operations • Keep operational area cool	2	1	2	Group 5 Safety Officer (Logan Smith), Before, During & After Operation	Ongoing

RISK MATRIX: (To generate the risk level).

Very likely 5	5	10	15	20	25
Likely 4	4	8	12	16	20
Possible 3	3	6	9	12	15
Unlikely 2	2	4	6	8	10
Extremely unlikely 1	1	2	3	4	5
Likelihood (L) ↑ / Severity (S) →	Minor injury – No first aid treatment required 1	Minor injury – Requires First Aid Treatment 2	Injury - requires GP treatment or Hospital attendance 3	Major Injury 4	Fatality 5

ACTION LEVEL: (To identify what action needs to be taken).

POINTS:	RISK LEVEL:	ACTION:
1 – 2	NEGLIGIBLE	No further action is necessary.
3 – 5	TOLERABLE	Where possible, reduce the risk further
6 - 12	MODERATE	Additional control measures are required
15 – 16	HIGH	Immediate action is necessary
20 - 25	INTOLERABLE	Stop the activity/ do not start the activity